

*Abstract Abstract goes here.

Introduction

Background Lung disease remains a leading cause of global morbidity and mortality. In the diagnosis of lung diseases

Computed Tomography (CT) is a radiographic technology that combines X-ray equipment with computer reconstruction

A defining characteristic of a CT scan is the Hounsfield Unit (HU), a quantitative scale of radiodensity that allows

The traditional approach for CT analysis relies on a "slice-by-slice" approach where a radiologist views a series of axial

3D volumetric analysis represents a quantitative shift wherein the entire dataset is used to reconstruct the lung's anatomy

Modern radiology increasingly relies on Artificial Intelligence (AI) to streamline the acquisition of radiologist analysis

Problem Statement Modern radiology increasingly relies on AI to manage the vast quantities of 3D data produced by

Computational Overhead Processing a 3D CT scan involves treating stacked 2D slices as a single volumetric matrix

Interpretability Bottleneck Beyond computational limits, the clinical utility of deep learning is hampered by the interpretability

Objectives and Scope Objectives

Develop a Dictionary Learning framework for Interpretable Lung Disease Classification from 3D Chest CT

Implement training in two phases for normal and pathological atoms

Integrate KSVD with label consistency constraints

Visualize dictionary atoms and map them to anatomical features

Compare sparse coefficient patterns across disease classes

Visualize dictionary atoms using an interactive 3D interface

Demonstrate atom-to-pathology correspondence

Benchmark against baseline models

Compare with standard KSVD without label consistency constraint

Compare with 3D CNN baseline (3D-ResNet)

Integrate clinical metrics for volumetric analysis

Implement HU-based segmentation for GGO and consolidation

Calculate C/G ratio and compare model predictions against clinical thresholds

Scope

The proposed solution is limited to volumetric CT scans of the thorax and does not consider other imaging modalities such as MRI

The methodology is restricted to dictionary learning-based approaches, with particular emphasis on KSVD and LC-KSVD

Clinically, the analysis is confined to the characterization of lung parenchymal abnormalities, specifically GGO and consolidation

The scope excludes real-time clinical deployment, cross-institutional generalization studies, and regulatory validation. The

Literature Review

Previous Work

Deep Learning for Lung Disease Diagnosis from Medical Imaging The field of automated lung disease diagnosis has

Image Segmentation and Classification

Advancing Diagnostics with AI and CAD Systems

Prognostics with Radiomics and Predictive Analytics

Workflow Optimisation Using AI

Convolutional Neural Networks (CNNs) have emerged as the dominant paradigm for automated analysis of chest co

Sparse Dictionary Learning in Medical Image Analysis Sparse dictionary learning represents an unsupervised machi

The strength of dictionary learning lies in its ability to extract local, structural information from noisy medical ima

The success of these models hinges on the quality of the learned dictionary. Traditional methods like K-SVD focus

The LC-KSVD algorithm enhances the traditional K-SVD by adding a label consistency constraint and a classificat

Explainable AI in Medical Imaging Clinical radiology requires an understandable method for determining how the c

Predominant XAI approaches in medical imaging include:

Visual explanation techniques like Integrated Gradients and Gradient-weighted Class Activation Mapping (Grad-CAM)

LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive explanations) are two feature a

Attention mechanisms that provide more comprehensible decision logic by allowing models to selectively focus on pertin

According to studies, these explainability techniques greatly increase clinician comprehension and confidence in AI-

Volumetric Feature Extraction The core of pulmonary diagnostic analysis lies in the identification and quantificatio

Volumetric analysis allows for the quantification of these features beyond simple 2D identification. By segmenting t

Gaps in Literature

Integration of interpretability with 3D volumetric analysis is limited. While 3D CNNs and volumetric analysis meth

Existing post-hoc explanation methods for 3D data (e.g., 3D Grad-CAM) provide visualization of important region

Lack of Inherently Interpretable Deep Learning Architectures Modern XAI methodologies for medical imaging frequ

Sparse dictionary learning presents an alternative approach where learned dictionary elements can represent interpr

Frozen Dictionary Learning as a Mechanism for Pathological Differentiation A primary innovation of the proposed

The first phase involves learning a base dictionary using only "normal" or healthy lung data. These atoms capture

This method provides an unprecedented level of interpretability. When a query 3D patch is analyzed, the resulting

Handling Class Imbalance with Interpretable Models Rare lung pathologies—certain cancer subtypes (e.g., large cel

However, the application of class imbalance mitigation techniques within interpretable sparse representation framew

Methodology The research consists of main phases

Overall methodology Diagram

Data acquisition and pre-processing We selected two publicly available chest CT datasets, CT-RATE[citation] and

Data pre-processing corrections such as intensity normalization and spacing adjustments have already been implemen

Model Training

Model training will be implemented using a two-phase frozen dictionary learning approach to capture both normal

Phase 1: Learning the Normal Dictionary We will first train a dictionary using only healthy or normal samples from

Phase 2: Learning Pathological Atoms After the normal dictionary is established and fixed, we will train additional

We extend the dictionary learning approach using LC-KSVD to improve the discriminative power of the learned sp

We will formulate LC-KSVD to jointly optimize three key objectives:

Accurate reconstruction of CT image patches using the learned dictionary.

Consistency of sparse representations within the same disease class.

Effective classification of lung conditions based on the sparse codes.

Appropriate weighting will be applied to balance these objectives, and hyperparameters will be selected through validat

Classification and Disease Diagnosis Support Vector Machine (SVM) use as the final disease classification model be

SVMs are also known to be robust when dealing with limited and imbalanced medical data, making them a suitabl

The model will be implemented using the scikit-learn library in Python and will receive the sparse codes generated

The classifier will predict multiple disease categories, with the number of output classes matching the labeling struc

Interpretability Analysis and Visualization To improve interpretability and clinical relevance, the learned dictionary

Volumetric Feature Extraction and Quantification Ground-glass opacities (GGO) will be identified based on their c

Baseline Comparisons and Benchmarking To demonstrate the effectiveness of the proposed method, it will be comp

Evaluation Metrics and Validation Strategy The performance of the proposed model will be evaluated using accurac

Timeline and Resource Required

Timeline Provide a realistic timeline for completing the project. Use a Gantt chart or table.

Resource Required State the resource required for the project and estimate the budget for the resources required.

Conclusion Summarize the key points of your proposal and reiterate the importance of the project.